

10. (a) All three dice show even numbers if and only if all three of A , B , and C occur. So, the event is $A \cap B \cap C$.
- (b) None of the three dice show even numbers if and only if all three of A^c , B^c , and C^c occur. So, the event is $A^c \cap B^c \cap C^c$.
- (c) At least one die shows an odd number if and only if at least one of A^c , B^c , and C^c occur. So, the event is $A^c \cup B^c \cup C^c$.
- (d) At most two dice show odd numbers if and only if at least one die shows an even number, so the event is $A \cup B \cup C$. This can also be expressed as the union of the three events of the form $A \cap B \cap C^c$ where exactly one die shows odd together with the three events of the form $A \cap B^c \cap C^c$ where exactly two dice show odd together with the even $A \cap B \cap C$ where no dice show odd.
- (e) We can enumerate all the sums that are no greater than 5: $1 + 1 + 1$, $2 + 1 + 1$, $1 + 2 + 1$, $1 + 1 + 2$, $2 + 2 + 1$, $2 + 1 + 2$, and $1 + 2 + 2$. The first of these corresponds to the event $A_1 \cap B_1 \cap C_1$, the second to $A_2 \cap B_1 \cap C_1$, etc. The union of the seven such events is what is requested, namely

$$(A_1 \cap B_1 \cap C_1) \cup (A_2 \cap B_1 \cap C_1) \cup (A_1 \cap B_2 \cap C_1) \cup (A_1 \cap B_1 \cap C_2) \cup (A_2 \cap B_2 \cap C_1) \cup (A_2 \cap B_1 \cap C_2) \cup (A_1 \cap B_2 \cap C_2).$$

11. (a) All of the events mentioned can be determined by knowing the voltages of the two subcells. Hence the following set can serve as a sample space

$$S = \{(x, y) : 0 \leq x \leq 5 \text{ and } 0 \leq y \leq 5\},$$

where the first coordinate is the voltage of the first subcell and the second coordinate is the voltage of the second subcell. Any more complicated set from which these two voltages can be determined could serve as the sample space, so long as each outcome could at least hypothetically be learned.

- (b) The power cell is functional if and only if the sum of the voltages is at least 6. Hence, $A = \{(x, y) \in S : x + y \geq 6\}$. It is clear that $B = \{(x, y) \in S : x = y\}$ and $C = \{(x, y) \in S : x > y\}$. The powercell is not functional if and only if the sum of the voltages is less than 6. It needs less than one volt to be functional if and only if the sum of the voltages is greater than 5. The intersection of these two is the event $D = \{(x, y) \in S : 5 < x + y < 6\}$. The restriction “ $\in S$ ” that appears in each of these descriptions guarantees that the set is a subset of S . One could leave off this restriction and add the two restrictions $0 \leq x \leq 5$ and $0 \leq y \leq 5$ to each set.
- (c) The description can be worded as “the power cell is not functional, and needs at least one more volt to be functional, and both subcells have the same voltage.” This is the intersection of A^c , D^c , and B . That is, $A^c \cap D^c \cap B$. The part of D^c in which $x + y \geq 6$ is not part of this set because of the intersection with A^c .
- (d) We need the intersection of A^c (not functional) with C^c (second subcell at least as big as first) and with B^c (subcells are not the same). In particular, $C^c \cap B^c$ is the event that the second subcell is strictly higher than the first. So, the event is $A^c \cap B^c \cap C^c$.

1.5 The Definition of Probability

Solutions to Exercises

1. Define the following events:

$$\begin{aligned} A &= \{\text{the selected ball is red}\}, \\ B &= \{\text{the selected ball is white}\}, \\ C &= \{\text{the selected ball is either blue, yellow, or green}\}. \end{aligned}$$

We are asked to find $\Pr(C)$. The three events A , B , and C are disjoint and $A \cup B \cup C = S$. So $1 = \Pr(A) + \Pr(B) + \Pr(C)$. We are told that $\Pr(A) = 1/5$ and $\Pr(B) = 2/5$. It follows that $\Pr(C) = 2/5$.

2. Let B be the event that a boy is selected, and let G be the event that a girl is selected. We are told that $B \cup G = S$, so $G = B^c$. Since $\Pr(B) = 0.3$, it follows that $\Pr(G) = 0.7$.
3. (a) If A and B are disjoint then $B \subset A^c$ and $BA^c = B$, so $\Pr(BA^c) = \Pr(B) = 1/2$.
 (b) If $A \subset B$, then $B = A \cup (BA^c)$ with A and BA^c disjoint. So $\Pr(B) = \Pr(A) + \Pr(BA^c)$. That is, $1/2 = 1/3 + \Pr(BA^c)$, so $\Pr(BA^c) = 1/6$.
 (c) According to Theorem 1.4.11, $B = (BA) \cup (BA^c)$. Also, BA and BA^c are disjoint so, $\Pr(B) = \Pr(BA) + \Pr(BA^c)$. That is, $1/2 = 1/8 + \Pr(BA^c)$, so $\Pr(BA^c) = 3/8$.
4. Let E_1 be the event that student A fails and let E_2 be the event that student B fails. We want $\Pr(E_1 \cup E_2)$. We are told that $\Pr(E_1) = 0.5$, $\Pr(E_2) = 0.2$, and $\Pr(E_1 E_2) = 0.1$. According to Theorem 1.5.7, $\Pr(E_1 \cup E_2) = 0.5 + 0.2 - 0.1 = 0.6$.
5. Using the same notation as in Exercise 4, we now want $\Pr(E_1^c \cap E_2^c)$. According to Theorems 1.4.9 and 1.5.3, this equals $1 - \Pr(E_1 \cup E_2) = 0.4$.
6. Using the same notation as in Exercise 4, we now want $\Pr([E_1 \cap E_2^c] \cup [E_1^c \cap E_2])$. These two events are disjoint, so

$$\Pr([E_1 \cap E_2^c] \cup [E_1^c \cap E_2]) = \Pr(E_1 \cap E_2^c) + \Pr(E_1^c \cap E_2).$$

Use the reasoning from part (c) of Exercise 3 above to conclude that

$$\begin{aligned}\Pr(E_1 \cap E_2^c) &= \Pr(E_1) - \Pr(E_1 \cap E_2) = 0.4, \\ \Pr(E_1^c \cap E_2) &= \Pr(E_2) - \Pr(E_1 \cap E_2) = 0.1.\end{aligned}$$

It follows that the probability we want is 0.5.

7. Rearranging terms in Eq. (1.5.1) of the text, we get

$$\Pr(A \cap B) = \Pr(A) + \Pr(B) - \Pr(A \cup B) = 0.4 + 0.7 - \Pr(A \cup B) = 1.1 - \Pr(A \cup B).$$

So $\Pr(A \cap B)$ is largest when $\Pr(A \cup B)$ is smallest and vice-versa. The smallest possible value for $\Pr(A \cup B)$ occurs when one of the events is a subset of the other. In the present exercise this could only happen if $A \subset B$, in which case $\Pr(A \cup B) = \Pr(B) = 0.7$, and $\Pr(A \cap B) = 0.4$. The largest possible value of $\Pr(A \cup B)$ occurs when either A and B are disjoint or when $A \cup B = S$. The former is not possible since the probabilities are too large, but the latter is possible. In this case $\Pr(A \cup B) = 1$ and $\Pr(A \cap B) = 0.1$.

8. Let A be the event that a randomly selected family subscribes to the morning paper, and let B be the event that a randomly selected family subscribes to the afternoon paper. We are told that $\Pr(A) = 0.5$, $\Pr(B) = 0.65$, and $\Pr(A \cup B) = 0.85$. We are asked to find $\Pr(A \cap B)$. Using Theorem 1.5.7 in the text we obtain

$$\Pr(A \cap B) = \Pr(A) + \Pr(B) - \Pr(A \cup B) = 0.5 + 0.65 - 0.85 = 0.3.$$

9. The required probability is

$$\begin{aligned}\Pr(A \cap B^C) + \Pr(A^C B) &= [\Pr(A) - \Pr(A \cap B)] + [\Pr(B) - \Pr(A \cap B)] \\ &= \Pr(A) + \Pr(B) - 2\Pr(A \cap B).\end{aligned}$$

10. Theorem 1.4.11 says that $A = (A \cap B) \cup (A \cap B^c)$. Clearly the two events $A \cap B$ and $A \cap B^c$ are disjoint. It follows from Theorem 1.5.6 that $\Pr(A) = \Pr(A \cap B) + \Pr(A \cap B^c)$.

11. (a) The set of points for which $(x - 1/2)^2 + (y - 1/2)^2 < 1/4$ is the interior of a circle that is contained in the unit square. (Its center is $(1/2, 1/2)$ and its radius is $1/2$.) The area of this circle is $\pi/4$, so the area of the remaining region (what we want) is $1 - \pi/4$.

(b) We need the area of the region between the two lines $y = 1/2 - x$ and $y = 3/2 - x$. The remaining area is the union of two right triangles with base and height both equal to $1/2$. Each triangle has area $1/8$, so the region between the two lines has area $1 - 2/8 = 3/4$.

(c) We can use calculus to do this. We want the area under the curve $y = 1 - x^2$ between $x = 0$ and $x = 1$. This equals

$$\int_0^1 (1 - x^2) dx = x - \frac{x^3}{3} \Big|_{x=0}^1 = \frac{2}{3}.$$

(d) The area of a line is 0, so the probability of a line segment is 0.

12. The events $B_1, B_2, \dots$ are disjoint, because the event B_1 contains the points in A_1 , the event B_2 contains the points in A_2 but not in A_1 , the event B_3 contains the points in A_3 but not in A_1 or A_2 , etc. By this same reasoning, it is seen that $\cup_{i=1}^n A_i = \cup_{i=1}^n B_i$ and $\cup_{i=1}^\infty A_i = \cup_{i=1}^\infty B_i$. Therefore,

$$\Pr\left(\bigcup_{i=1}^n A_i\right) = \Pr\left(\bigcup_{i=1}^n B_i\right)$$

and

$$\Pr\left(\bigcup_{i=1}^\infty A_i\right) = \Pr\left(\bigcup_{i=1}^\infty B_i\right).$$

However, since the events $B_1, B_2, \dots$ are disjoint,

$$\Pr\left(\bigcup_{i=1}^n B_i\right) = \sum_{i=1}^n \Pr(B_i)$$

and

$$\Pr\left(\bigcup_{i=1}^\infty B_i\right) = \sum_{i=1}^\infty \Pr(B_i).$$

13. We know from Exercise 12 that

$$\Pr\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \Pr(B_i).$$

Furthermore, from the definition of the events $B_1, \dots, B_n$ it is seen that $B_i \subset A_i$ for $i = 1, \dots, n$. Therefore, by Theorem 1.5.4, $\Pr(B_i) \leq \Pr(A_i)$ for $i = 1, \dots, n$. It now follows that

$$\Pr\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n \Pr(A_i).$$

(Of course, if the events $A_1, \dots, A_n$ are disjoint, there is equality in this relation.)

For the second part, apply the first part with A_i replaced by A_i^c for $i = 1, \dots, n$. We get

$$\Pr\left(\bigcup_{i=1}^n A_i^c\right) \leq \sum_{i=1}^n \Pr(A_i^c). \quad (\text{S.1.1})$$

Exercise 5 in Sec. 1.4 says that the left side of (S.1.1) is $\Pr([\bigcap A_i]^c)$. Theorem 1.5.3 says that this last probability is $1 - \Pr(\bigcap A_i)$. Hence, we can rewrite (S.1.1) as

$$1 - \Pr\left(\bigcap_{i=1}^n A_i\right) \leq \sum_{i=1}^n \Pr(A_i^c).$$

Finally take one minus both sides of the above inequality (which reverses the inequality) and produces the desired result.

14. First, note that the probability of type AB blood is $1 - (0.5 + 0.34 + 0.12) = 0.04$ by using Theorems 1.5.2 and 1.5.3.
 - (a) The probability of blood reacting to anti-A is the probability that the blood is either type A or type AB. Since these are disjoint events, the probability is the sum of the two probabilities, namely $0.34 + 0.04 = 0.38$. Similarly, the probability of reacting with anti-B is the probability of being either type B or type AB, $0.12 + 0.04 = 0.16$.
 - (b) The probability that both antigens react is the probability of type AB blood, namely 0.04.

1.6 Finite Sample Spaces

Solutions to Exercises

1. The safe way to obtain the answer at this stage of our development is to count that 18 of the 36 outcomes in the sample space yield an odd sum. Another way to solve the problem is to note that regardless of what number appears on the first die, there are three numbers on the second die that will yield an odd sum and three numbers that will yield an even sum. Either way the probability is $1/2$.
2. The event whose probability we want is the complement of the event in Exercise 1, so the probability is also $1/2$.
3. The only differences greater than or equal to 3 that are available are 3, 4 and 5. These large difference only occur for the six outcomes in the upper right and the six outcomes in the lower left of the array in Example 1.6.5 of the text. So the probability we want is $1 - 12/36 = 2/3$.
4. Let x be the proportion of the school in grade 3 (the same as grades 2–6). Then $2x$ is the proportion in grade 1 and $1 = 2x + 5x = 7x$. So $x = 1/7$, which is the probability that a randomly selected student will be in grade 3.